

Dear Hiring Manager,

I am writing to express my interest in the entry-level Quantitative Consultant role within the Clearing Department at CME. My educational background in Physics and Risk Management and Financial Engineering provides the analytical and engineering skills needed to excel in this role. Additionally, my work experience as a researcher, running high-performance Monte Carlo simulations for benchmarking and validating physical models, fits well with the associated responsibilities.

CME is an inspiring institution for me because it is arguably the most radical innovator in financial derivatives and the way they are traded. An entry-level position in the Clearing House department would provide me with an objective view of how the market works, which aligns with my long-term career goals. I am interested in pursuing a career in quantitative risk management because financial crises are among the most interesting phenomena in finance and in general. Events that seem obvious in retrospect often flew under the radar throughout history. Contributing to the understanding of these risks and crises is a meaningful pursuit. The SPAN 2 system presents unique challenges in high-performance, low-latency system design and programming due to the sheer volume of data and the complexity of the numerous models it encompasses; this is exactly the kind of challenge I am passionate about taking on.

My education and work experience in Physics provided me with strong skills in conducting independent first-principles research effectively and the ability to clearly communicate results to stakeholders across a diverse range of expertise. Specifically, the skills I built working on large-scale and high-performance Monte Carlo simulations for model validation are directly transferrable to this role. My M.Sc. degree from the Risk Management and Financial Engineering program from Imperial College London equipped me with the knowledge of stochastic calculus, financial markets, and quantitative risk models required for this position. As a side project, I built concurrent data pipelines streaming and logging high-frequency market data from exchange APIs on AWS EC2 servers. This gave me hands-on experience in low-latency programming with C++, distributed system design, and test-driven development best practices. Coming from Physics, I value data hygiene and explainable insights. The Newtonian conviction that anything can be modeled and understood is a triumph of the human spirit and what gets me out of bed in the morning.

Thank you for considering my application. I am excited about the opportunity to contribute to the team that manages the stability of global markets through frameworks like SPAN 2. I would appreciate the chance to discuss how my C++ proficiency and analytical mindset align with your team's needs in an interview.

Best Regards,
Mike Chen